Aadarsh Singh

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Research Interests

Flavour and neutrino physics. Fermion mass and mixing from theory-space geometry: clockwork, deconstructed and fractal mass chains; disorder and localization; rank-preserving control of chiral zero modes. Neutrino mass models and phenomenology; geometric phases in neutrino oscillations; neutrino–dark sector connections.

Effective field theories. Four-lepton SMEFT operators; perturbative unitarity, positivity and renormalization-group running; low-energy effective descriptions of leptonic operators; neutrino self-interactions.

Computational and AI methods for high-energy physics. Multi-field vacuum tunnelling and bounce solvers; benchmarking language models on physics software; agent-based and schema-mediated workflows; validation pipelines for collider simulation.

Quantum information and quantum computing (additional interest). Foundational aspects of quantum correlations; entanglement and information-theoretic measures in particle physics; quantum simulation of field theories.

Education

Indian Institute of Science (IISc), Bangalore *Oct 2020 – Present*
Ph.D. in Physics, Centre for High Energy Physics
Advisor: Prof. Sudhir K. Vempati
Expected defense: *Nov 2026*

IISER Bhopal *Aug 2015 – May 2020*
BS–MS (Integrated M.Sc.) in Physics, Minor in Mathematics
CPI: 9.3/10.0

Publications

Peer-Reviewed Articles

1. **A. Singh**, G. D'Ambrosio, S. K. Vempati, *Theoretical and Experimental Constraints in the μ–τ Four-Lepton Sector of the SMEFT: Implications to Neutrino Self-Interactions*, *Phys. Rev. D* **114**, 035038 (2026), DOI:10.1103/8z45-8cw5, arXiv:2604.27716 [hep-ph].
Derived theoretical and experimental constraints on dimension-six four-lepton SMEFT operators — perturbative unitarity, positivity, RG running and collider/low-energy bounds — and mapped the surviving parameter space for neutrino self-interactions.

2. **A. Singh**, *Independent Chiral Control in Theory-Space Models: A Rank-Preserving Framework and Its Application to Neutrino Mass Generation*, *Eur. Phys. J. C* **86**, 743 (2026), DOI:10.1140/epjc/s10052-026-15918-3, arXiv:2409.09033 [math.RA, hep-ph].
Established a general rank-preserving framework for independently controlling chiral zero-mode profiles in theory-space mass matrices, with applications to clockwork, deconstruction, Kaluza–Klein spectra and neutrino mass generation.

3. A. Ibarra, **A. Singh**, S. K. Vempati, *Flavour from Fractal Mass Chains*, *Phys. Lett. B* **874** (2026) 140268, DOI:10.1016/j.physletb.2026.140268, arXiv:2509.04811 [hep-ph].
Introduced fractal-inspired mass-matrix structures in which self-similar theory-space geometry generates three light fermionic modes and reproduces the observed lepton mass hierarchies and mixing patterns from few parameters.

4. **A. Singh**, *Revisiting Neutrino Masses in Clockwork Models*, *Eur. Phys. J. C* **85**, 655 (2025), DOI:10.1140/epjc/s10052-025-14370-z, arXiv:2407.13733 [hep-ph].
Built a systematic framework for neutrino mass generation in clockwork theory, resolving inconsistencies in earlier treatments and identifying viable parameter space for realistic phenomenology.

Preprints and Submitted Articles

5. **A. Singh**, S. K. Vempati, *HEPToolBench 1.2: Testing How Reliably Language Models Can Drive Particle Physics Software*, arXiv:2608.28232 [hep-ph] (2026). Submitted to *Computer Physics Communications*.
Introduced a benchmark of 28 collider-simulation tasks measuring how reliably language models produce executable instructions for standard HEP software, and showed that a schema-mediated interface raises mean performance from 0.418 to 0.902 across 42 locally-served and frontier model deployments.

6. **A. Singh**, S. K. Vempati, *HEPLocalAgent 1.0: Running Collider Simulations from Plain-Language Requests on Your Own Computer*, arXiv:2608.28244 [hep-ph] (2026). Submitted to *Computer Physics Communications*.
Built an open-source system that turns plain-language requests into validated collider-simulation workflows using a locally served language model with deterministic software validation, raising the number of benchmark-conforming artifacts from 11 to 43 across the pipeline.

7. S. R. Kannagi, **A. Singh**, S. K. Vempati, *Fourier-Preconditioned Path Deformations for Multi-Field Vacuum Tunnelling*, arXiv:2606.17485 [hep-ph] (2026). Submitted to *JCAP*; referee report received, revision in preparation.
Developed an endpoint-safe Fourier representation for multi-field tunnelling paths, usable both as a variational ansatz and as a preconditioner for numerical bounce solvers; benchmarked in systems of up to 50 scalar fields.

8. **A. Singh**, S. K. Vempati, *Can Randomness Lead to Non-Anarchical Mixing Angles?*, arXiv:2512.21202 [hep-ph] (2025). Submitted to *PRD*.
Analyzed disordered fermion theory spaces across nearest-neighbour, fully connected and Petersen-graph geometries, identifying a weak-disorder regime in which geometry-dependent localization yields realistic mass hierarchies together with non-anarchical mixing.

Conference Proceedings

9. S. K. Vempati, **A. Singh**, *Neutrino Masses and Mixing Driven by Randomness*, *PoS* **HQL2023** (2024) 025, DOI:10.22323/1.462.0025.

10. **A. Singh**, S. K. Vempati, *Fermion Masses from Structures in Abstract Spaces*, *Proc. XXV DAE-BRNS High Energy Physics Symposium 2022*, Springer Proc. Phys. **304** (2024) 330–333, DOI:10.1007/978-981-97-0289-3_72.

Software

11. **A. Singh**, S. K. Vempati, *HEPToolBench*: a deterministic benchmark for evaluating language models on high-energy-physics software workflows. Version 1.2, Zenodo (2026). DOI:10.5281/zenodo.21903075. Code: github.com/AadarshSingh0/HEPToolBench.

12. **A. Singh**, S. K. Vempati, *HEPLocalAgent*: an open-source system for running validated collider-simulation workflows from plain-language requests on local hardware. Version 1.0.1, Zenodo (2026). DOI:10.5281/zenodo.21904178. Code: github.com/AadarshSingh0/HEPLocalAgent.

13. **A. Singh**, *Fourier Path Bounce*: an open-source Python/Wolfram package for endpoint-safe Fourier preconditioning of multi-field vacuum-tunnelling paths, with configurable mode optimization, a command-line interface, and interfaces to CosmoTransitions and FindBounce. Version 0.1.0, Zenodo (2026). DOI:10.5281/zenodo.22294296. Code: github.com/AadarshSingh0/fourier-path-bounce.

Public repositories with the code and data reproducing results from individual papers are linked from the GitHub profile above.

In Preparation

- A. Dighe, **A. Singh**, S. K. Vempati, *Geometric phases in neutrino oscillations with heavy neutral leptons: phenomenological signatures* (in preparation).
- A. Ibarra, **A. Singh**, A. Srivatsa, S. K. Vempati, *A full flavour model for quarks and leptons using fractal chains* (in preparation).

Research Experience

Doctoral Research, Centre for High Energy Physics, IISc Bangalore *2020 – Present*
Advisor: Prof. Sudhir K. Vempati

- Constructed and analyzed BSM frameworks for fermion mass generation using deconstruction, clockwork and fractal theory-space models, showing how geometric and self-similar structures generate the observed mass hierarchies and mixing patterns.
- Developed a general framework of rank-preserving transformations for structured mass matrices, enabling independent control of chiral zero-mode profiles across theory-space models.
- Studied stochastic and disordered fermion theory spaces and the role of geometry-dependent localization in producing hierarchical masses and structured flavor mixing.
- Derived theoretical and experimental constraints on four-lepton SMEFT operators and their implications for neutrino self-interactions and leptophilic new physics, in collaboration with G. D'Ambrosio (INFN Napoli).
- Developed and benchmarked Fourier-based path-deformation methods for multi-field vacuum tunnelling and numerical bounce calculations.
- Built and benchmarked agent-based systems that drive standard HEP software (FeynRules, MadGraph, Pythia, Delphes) from natural-language specifications, including a 28-task benchmark for evaluating language-model reliability on collider-simulation workflows.
- Ongoing: neutrino–dark matter interactions and flavor-dependent signatures in high-energy cosmic neutrinos; geometric phase effects in neutrino systems with heavy neutral leptons.

M.S. Thesis, IISER Bhopal *2019 – 2020*
Symmetries of asymptotically flat spacetimes

- Studied Bondi–Metzner–Sachs (BMS) symmetries and their generalizations under vector-field extensions, analyzing extended algebraic structures in the asymptotically flat limit.

Research Internships

- **Indian Statistical Institute, Kolkata** *Summer 2017*
  Quantum information theory and foundational aspects of quantum correlations.
- **Harish-Chandra Research Institute, Prayagraj** *Summer 2018*
  Summer Programme in Mathematics (SPIM): advanced mathematical methods.

Talks and Presentations

Invited Talks and Seminars

- *Flavours from Fractals*, INFN Sezione di Napoli and Università di Napoli Federico II, Naples, Italy (Nov 2025).
- *Flavours from Fractals*, School of Physics, University of New South Wales, Sydney, Australia (Mar 2025).

Contributed Talks

- *Can Randomness Lead to Non-Anarchical Mixing Angles?* (parallel talk), PLANCK2026 and 6th EuCAPT Symposium, CERN, Geneva, Switzerland (18–22 May 2026).
- *Can Hierarchy and Generation Have the Same Solution?*, Theory Meets Experiments 2025, Galileo Galilei Institute, Florence, Italy (Nov 2025).
- *Are Fractals Behind Flavor Mixing of the SM?*, KEK-PH 2025 Winter Workshop, KEK, Tsukuba, Japan (Feb 2025).
- *Are Fractals Behind Flavour Mixing of the SM?*, BCVSPIN 2024: Particle Physics and Cosmology in the Himalayas, Kathmandu, Nepal (9–13 Dec 2024).
- *Fermion Masses from Structures in Abstract Spaces*, XXV DAE-BRNS High Energy Physics Symposium, IISER Mohali, India (12–16 Dec 2022).

Schools and Training Workshops

- HSF-India HEP Software and Scientific Computing Workshops, NISER Bhubaneswar (18–22 Dec 2023) and IISc Bangalore (15–21 Jun 2025).
- Iwate Collider School (IWS 2025), Iwate, Japan (Feb–Mar 2025) — presented *A Fractal Theory for Flavor*.
- Understanding the Universe Through Neutrinos (NEUS 2024), ICTS-TIFR, Bangalore (22 Apr – 3 May 2024).
- Statistical Methods and Machine Learning in High Energy Physics (ML4HEP), ICTS-TIFR, Bangalore (28 Aug – 8 Sep 2023).

Honors and Awards

- All-India Rank 11, CSIR-NET Physics (national research-fellowship qualifying examination, India), June 2019 — awarded CSIR Junior Research Fellowship.

- All-India Rank 16, CSIR-NET Physics, December 2018.
- All-India Rank 16, Joint Entrance Screening Test (JEST) Physics, 2020.
- All-India Rank 57, GATE Physics, 2020.

Technical Skills

HEP and phenomenology: FeynRules, MadGraph5_aMC@NLO, Pythia, Delphes, CosmoTransitions.

Programming and computation: Python (NumPy, SciPy, Matplotlib, JAX), Mathematica (FeynCalc, FeynArts, symbolic computation), C, MATLAB, SQL, Git, Linux/Unix, LaTeX.

AI-assisted scientific computing: LLM benchmarking and agent-based workflows for HEP; schema-mediated tool interfaces, structured-output and validation pipelines for collider-simulation tasks. Both systems are released as open-source software with archived Zenodo versions (see Software above).

Teaching and Mentoring

Mentoring

- Mentored three M.Sc. summer interns on a project in quantum computing, Dr NGP Arts and Science College, Coimbatore (2023).
- Mentored S. R. Kannagi (BS student, IISc) on Fourier-preconditioned bounce methods for multi-field vacuum tunnelling; the work led to arXiv:2606.17485.

Teaching Assistant, Indian Institute of Science, Bangalore

- Quantum Mechanics II *Jan–May 2023*
- Mathematical Methods for Physics *Aug–Dec 2023*
- Statistical Mechanics *Jan–May 2022*

Teaching Assistant, NPTEL (National Programme on Technology Enhanced Learning)

- Relativistic Quantum Mechanics *Jul–Oct 2025*
- Quantum Field Theory and Quantum Entanglement *Jul–Oct 2026*